Tedla (Teddy) Tesfaye Godebo

Ancona, Italy · Kigali, Rwanda · Open to remote and EU roles | tedlatesfaye1@gmail.com | LinkedIn | GitHub | Portfolio

PROFILE

Third-year Software Engineering student specialising in machine learning, focused on **speech recognition for under-resourced African languages**. Hands-on with Whisper fine-tuning under tight GPU budgets, audio and vision deep learning, and shipping models as APIs with Docker. Native speaker of Wolaita. Seeking ML/speech internships, part-time remote work, and roles from mid-2027.

EDUCATION

African Leadership University (ALU) — Kigali, Rwanda May 2024 – May 2027 (expected)

- BSc Software Engineering, Machine Learning specialisation · GPA 4.5 / 5.0

Università Politecnica delle Marche (UNIVPM) — Ancona, Italy 2025 – present

- BSc Digital Economics and Business (concurrent degree)

SPEECH & ML RESEARCH

Google WAXAL ASR Challenge (Zindi) Ongoing

- Fine-tuning Whisper for Shona, Lingala and Luganda speech recognition in a $10,000-prize competition.
- Ran full-data whisper-small fine-tuning on free T4 GPUs across Colab and Kaggle, solving VRAM limits, dtype mismatches and out-of-memory failures.

Whisper-Based ASR for Wolaita (Wolaytta) — independent research paper In progress

- Adapting Whisper to Wolaita, an under-resourced Ethiopian Omotic language and my native language.
- Assembled the data plan: a public Hugging Face corpus (14.2k transcribed clips, 109 speakers), prior AAU Wolaytta ASR research, and radio, TV and Bible.is audio for additional training and evaluation data.

Lecture Chapter Mapper Independent project

- Built a faster-whisper + Claude API tool that transcribes lecture recordings and generates a clickable HTML chapter map, tagging each segment as essential, useful or skippable.

SELECTED ML PROJECTS

Acoustic Anomaly Detection Pipeline (MLOps) ALU coursework

- End-to-end system on the MIMII industrial pump audio dataset: CNN classifier (93.6% accuracy), FastAPI backend, Streamlit UI, Docker deployment with nginx load balancing.
- Load-tested with Locust across 1, 3 and 5 containers; resolved Apple Silicon and Keras version-compatibility issues.

Malaria Cell Image Classification ALU coursework · group lead

- Led a team comparing Baseline, VGG16, ResNet50 and InceptionV3 CNNs on the NIH malaria dataset.
- Best individual model: partially fine-tuned ResNet50 (F1 0.777, AUC 0.871); debugged a teammate's InceptionV3 pipeline and co-authored the report.

Africa Infrastructure Cost Predictor Independent project

- Regression model on 4,658 World Bank African infrastructure projects (1970s–2019) reaching R^2 0.923; served as a documented FastAPI REST API.

Water Pump Functionality Prediction DrivenData / Taarifa dataset

- Tuned Random Forest pipeline for pump-status classification (test macro-F1 0.701), with an IEEE-style report.

SOFTWARE ENGINEERING

DataBric — peer-to-peer mobile data sharing app Independent, in progress

- Flutter Android app with a FastAPI backend, Supabase database and relay infrastructure; testing a European relay first.

Auto Commit Agent (VS Code extension) Independent project

- TypeScript extension using the Anthropic API to generate contextual git commit messages.

EXPERIENCE

Freelance Coding & Data Science Tutor — Malta-based clients Ongoing

- Designed and taught a three-session “Python for Data Basics” course (Pandas, data cleaning, EDA, Matplotlib) with full notebooks, exercises and solutions.

Expansion Coordinator — YeneHealth Sep 2024 – Apr 2025

- Coordinated two cross-functional intern teams (Marketing, Operations); wrote the handover report used for future intern onboarding.

Tilting Futures Scholar — CEMACS, Universiti Sains Malaysia, Penang Semester placement

- Data cleaning, analysis and visualisation for marine conservation projects.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, Dart
ML / DL: PyTorch, TensorFlow/Keras, scikit-learn, Hugging Face Transformers, Whisper / faster-whisper, XGBoost, CNNs (ResNet, VGG, Inception)
Engineering: FastAPI, Streamlit, Flutter, Docker, nginx, Locust, Supabase, MySQL, MongoDB, Git, Colab, Kaggle